

CS501 DBMS LECTURE NOTES - UNIT 3:

NORMALIZATION

Page 1: Functional Dependencies & Normal Forms

1NF (First Normal Form): Eliminates repeating groups and ensures atomic values in all attributes.

2NF (Second Normal Form): Requires 1NF and ensures no non-prime attribute is partially dependent on any candidate key.

3NF (Third Normal Form): Requires 2NF and ensures no non-prime attribute is transitively dependent on candidate keys.

BCNF (Boyce-Codd Normal Form): For every non-trivial functional dependency $X \rightarrow Y$, X must be a super key.